Learning Objectives:

- Understand the basic concepts of Decision Tree and Decision Tree Pruning
- Learn how to build Decision Trees in Python
- Gain perspective on the applicability of Decision Tree and how it can be used to solve a variety of problem

Structure of the Session

Duration	Topic	Details
5 Min	Explaining the session structure	• Discussion of the agenda and learning objectives of the session.
30 min	Discussion of videos of the week	• Use the Discussion Questions to gauge the level of understanding of the group on videos covered during the week and identify areas to focus on • Make the group understand the concept of a Decision Tree and pruning a decision tree where the group needs help
10 min	EDA	• Discussion of data cleaning, key observations, and insights from the EDA section.
60 min	Hands-on Case study	• Build a Decision Tree model on the dataset provided • Prune the model to address overfitting
10 min	Extended QnAs	• Discuss practical applications, exchange thoughts, and challenges around Decision Tree models
5 min	Summarize the session	• In simple bullet points, cover the contents covered during the session

Important points to note:

- In the hands-on section, it is important that the focus stays on the implementation of the Decision Tree and not too much on the EDA section as it covers the concepts which are similar to the concepts discussed multiple times previously. We have noticed that if the plan is not followed the session exceeds the time limit.
- Be well prepared for the session and ensure to watch all the videos, related codes, and datasets. (It renders the Mentored Learning session of no use if we are not completely prepared)
- Log in for your session 5 min before and make sure you are ready with all the material to be used for the session.